

Time Bank



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Project Overview

Time Bank is a web application for exchanging services through a virtual currency based on time credits. Users can publish services, request services offered by other users, manage received requests, chat with other users, check their purchase and sale history, review transactions, and recharge their wallet.

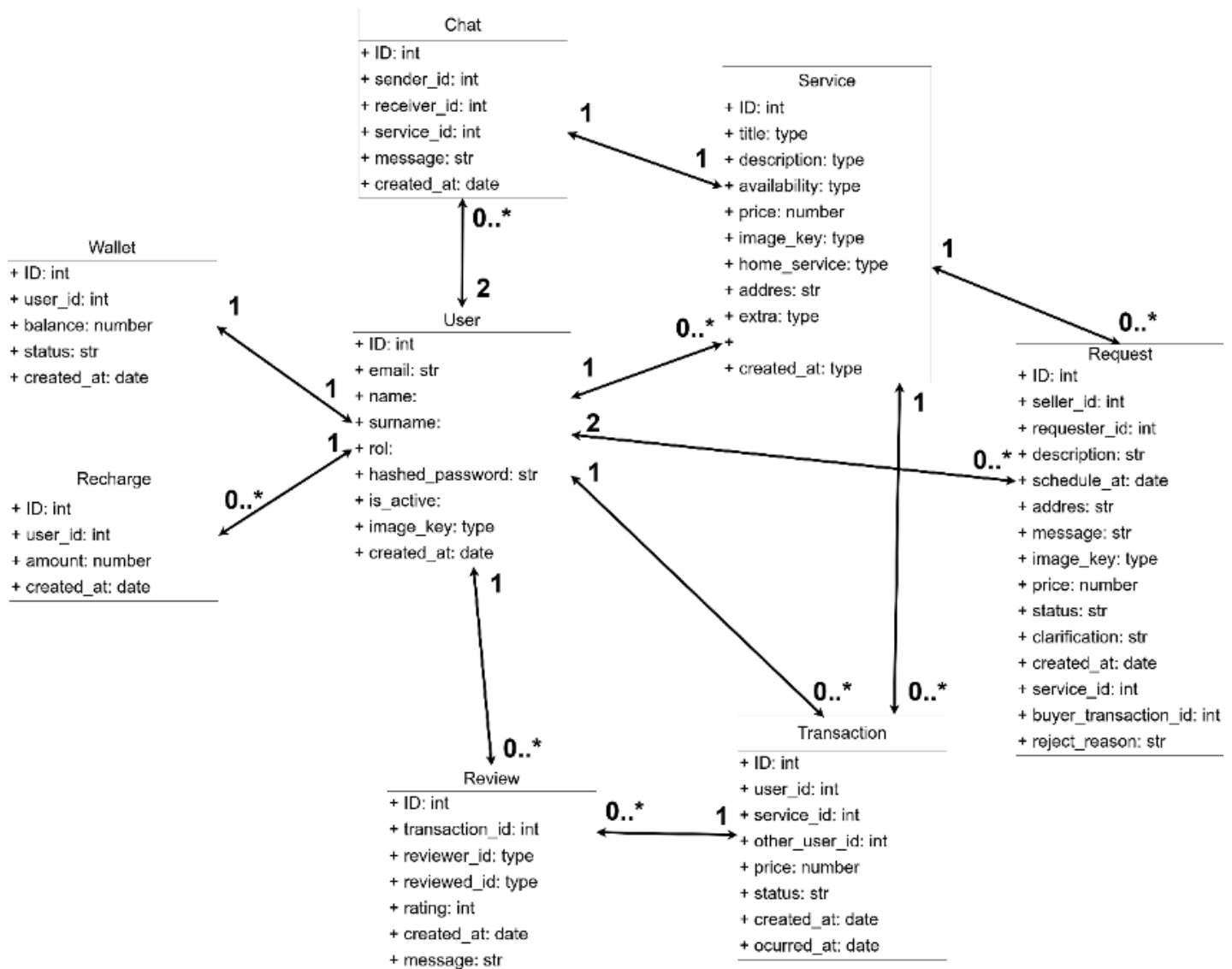
The system is divided into two repositories:

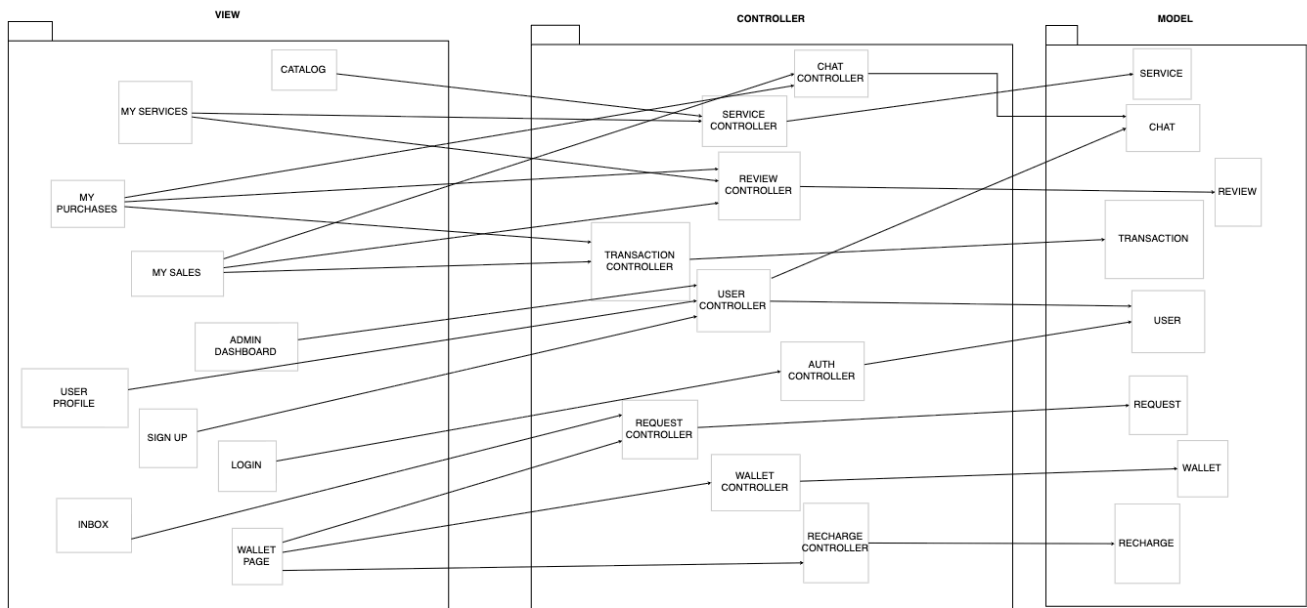
- **Frontend:** <https://github.com/carlotamorenouclm/TimeBankFront>
- **Backend:** <https://github.com/carlotamorenouclm/TimeBankBack>
- **Models:** https://drive.google.com/file/d/1-dKTxqYtD5g7gQ12-faBSI_ctbbNoa47/view?usp=sharing

Technology Stack

- **Frontend:** React, Vite, React Router, React Bootstrap, and Bootstrap.
- **Backend:** FastAPI, SQLAlchemy, Pydantic, and JWT authentication.
- **Database:** MySQL with PyMySQL.
- **Authentication:** OAuth2 Password Form with Bearer JWT tokens.
- **Payments:** Stripe Checkout and Stripe webhooks for wallet recharges.

Content Diagram





Main Entities

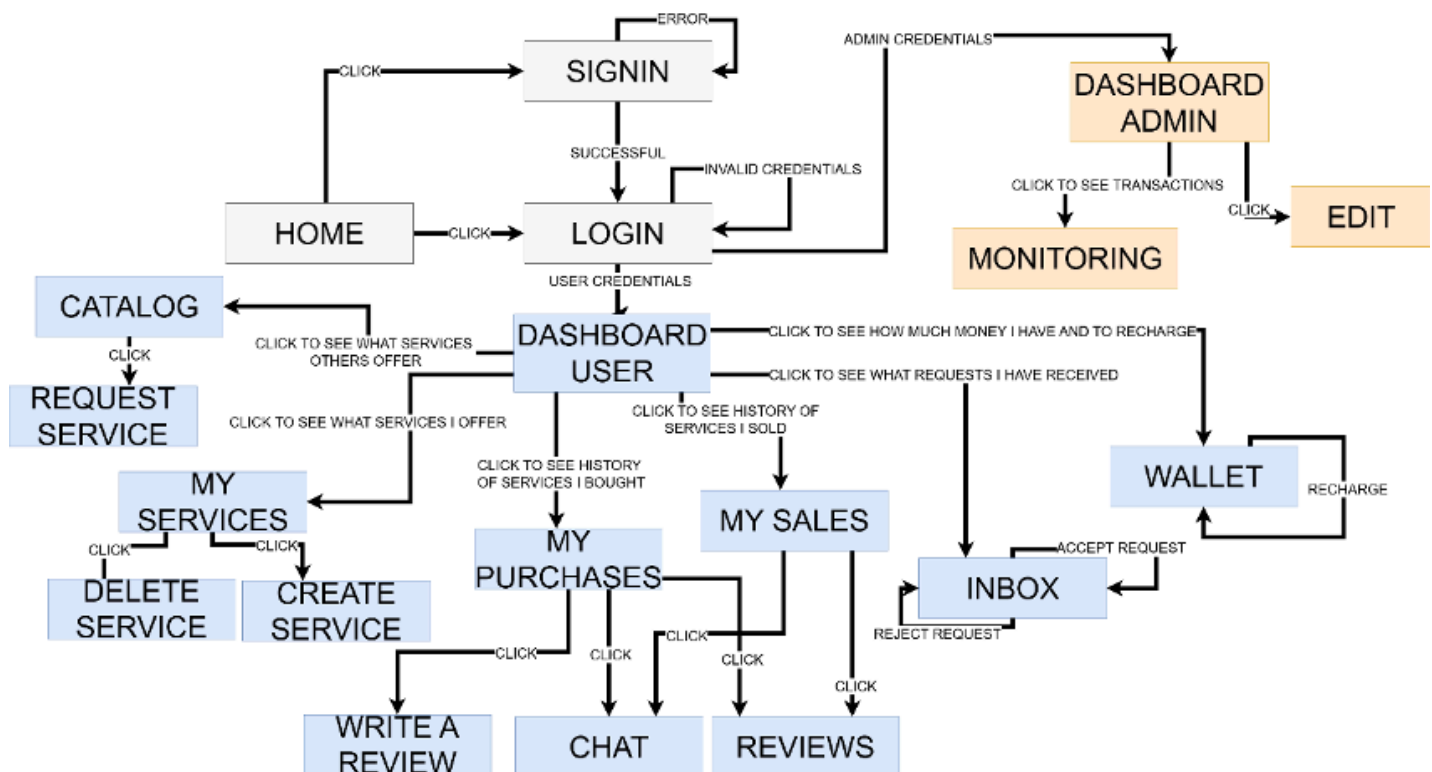
Entity	Main fields
User	id, email, hashed_password, name, surname, avatar_key, role, is_active, created_at
ServiceOffer	id, owner_id, title, description, availability, home_service, address, extra, price, image_key, owner_name, is_visible, created_at
ServiceRequest	id, receiver_id, requester_id, service_offer_id, buyer_transaction_id, requester_name, service, description, scheduled_at, address, message, image_key, price, status, clarification, reject_reason, created_at
UserWallet	id, user_id, balance, status, created_at
WalletRecharge	id, user_id, amount, created_at
UserTransaction	id, user_id, type, service, other_user, amount, status, occurred_at, created_at
ChatMessage	id, request_id, thread_key, sender_id, receiver_id, message, sent_at

Entity	Main fields
TransactionReview	id, transaction_id, reviewer_id, reviewed_user_id, rating, comment, created_at

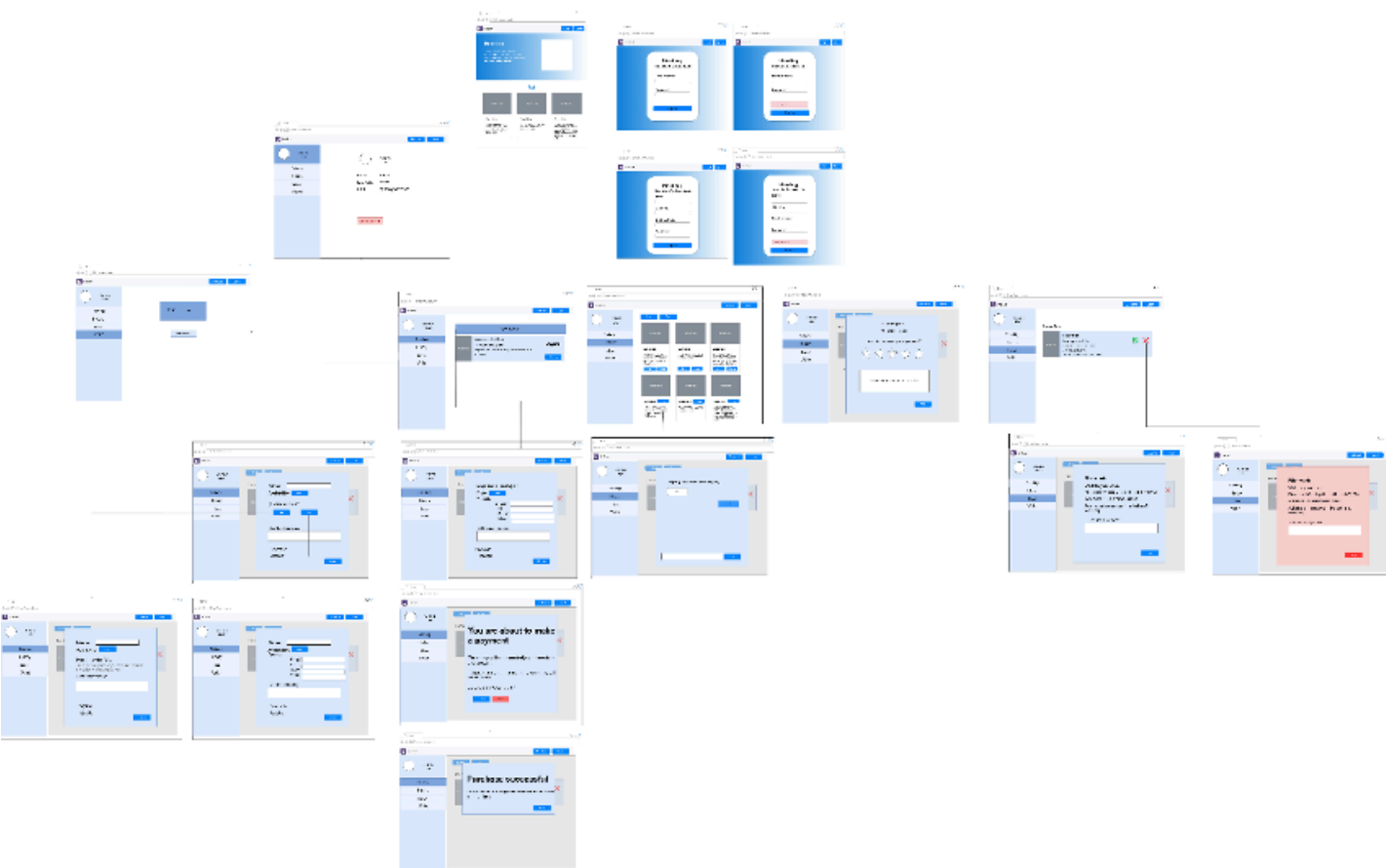
Main Relationships

- A user can publish zero or more services.
- A user has one associated wallet.
- A wallet can receive zero or more recharges.
- A user can have zero or more transactions.
- A user can send and receive zero or more service requests.
- A service offer can generate zero or more service requests.
- A transaction can be linked to a service request.
- A service request or thread can contain zero or more chat messages.
- A transaction can receive reviews related to the users involved.

Navigation Diagram



Presentation Diagram



API Dictionary

Health

Method	Endpoint	Authentication	Description	Request body	Response
GET	/health	No	Checks that the backend is available.	None	{ "status": "ok" }

Authenticated User

Method	Endpoint	Authentication	Description	Request body	Response
GET	/me	User	Returns the authenticated user's profile.	None	UserOut

Method	Endpoint	Authentication	Description	Request body	Response
POST	/me/update	User	Updates the authenticated user's profile.	email, name, surname, avatar_key	UserOut
POST	/me/password	User	Changes the authenticated user's password.	current_password, new_password	Text confirmation
DELETE	/me/delete	User	Deletes the authenticated user's account.	None	Empty 204 response
POST	/me/isAdmin	User	Indicates whether the authenticated user has the administrator role.	None	true or false

User Portal

Method	Endpoint	Authentication	Description	Request body	Response
GET	/portal/summary	User	Returns the summary displayed in the sidebar.	None	PortalUserSummary
GET	/portal/dashboard	User	Returns the available services and the user's own services.	None	DashboardResponse
GET	/portal/history	User	Returns the user's purchase and sale history.	None	HistoryResponse

Method	Endpoint	Authentication	Description	Request body	Response
POST	/portal/history/notifications/read	User	Marks purchase or sale notifications as read.	{ "transaction_type": "Purchase" } or { "transaction_type": "Sale" }	PortalUserSummary
GET	/portal/inbox	User	Returns received service requests.	None	InboxResponse
POST	/portal/inbox/{request_id}/accept	User	Accepts a received request.	{ "clarification": "... " }	Updated InboxResponse
POST	/portal/inbox/{request_id}/reject	User	Rejects a received request.	{ "reason": "..." }	Updated InboxResponse
POST	/portal/requests/{request_id}/complete	User	Completes an accepted request and updates the history.	None	HistoryResponse
GET	/portal/wallet	User	Returns wallet balance, status, and recharges.	None	WalletResponse
POST	/portal/wallet/recharge	User	Legacy manual recharge route. It currently rejects the operation and requires Stripe Checkout.	{ "amount": 10 }	Error 400
POST	/portal/wallet/checkout-session	User	Creates a Stripe Checkout session to recharge the wallet.	{ "amount": 10 }	StripeCheckoutSessionResponse

Method	Endpoint	Authentication	Description	Request body	Response
POST	/portal/wallet/checkout-session/{session_id}/confirm	User	Checks the status of a Checkout session and returns the wallet.	None	WalletResponse
POST	/portal/stripe/webhook	Stripe	Receives signed Stripe events and credits paid wallet recharges.	Stripe payload and stripe-signature header	{ "received": true }
POST	/portal/services/{service_offer_id}/request	User	Requests a service and deducts the payment from the wallet.	scheduled_at, street, street_number, floor, door, message	CreateServiceRequestResponse
POST	/portal/services	User	Publishes a new service offer.	title, description, availability, home_service, address, extra, price, image_key	CreateServiceOfferResponse
DELETE	/portal/services/{service_offer_id}	User	Deletes one of the user's own services.	None	DeleteServiceOfferResponse

Chat

Method	Endpoint	Authentication	Description	Request body	Response
GET	/chat/requests/{request_id}/messages	User	Lists the messages associated with a request.	None	ChatMessagesResponse
POST	/chat/requests/{request_id}/messages	User	Sends a message within a request.	{ "message": "... " }	ChatMessagesResponse

Method	Endpoint	Authentication	Description	Request body	Response
GET	/chat/threads/{thread_key}/messages	User	Lists the messages in a chat thread.	None	ChatMessagesResponse
POST	/chat/threads/{thread_key}/messages	User	Sends a message within a thread.	{ "receiver_id": 2, "message": "..."} }	ChatMessagesResponse

Reviews

Method	Endpoint	Authentication	Description	Request body	Response
POST	/reviews	User	Creates a review associated with a transaction.	transaction_id, rating, comment	CreateReviewResponse
GET	/reviews/services/{service_offer_id}	User	Returns the reviews for a service.	None	TransactionReviewOut[]
GET	/reviews/{transaction_id}	User	Returns the reviews for a transaction.	None	TransactionReviewOut[]

Users

Method	Endpoint	Authentication	Description	Request body	Response
POST	/users/signup	No	Creates a new user account.	email, password, name, surname	UserOut
GET	/users	Administrator	Lists all users.	None	UserOut[]
GET	/users/{user_id}	Administrator	Returns a user by identifier.	None	UserOut
POST	/users/update/{user_id}	Administrator	Updates another user's public data.	email, name, surname, avatar_key	Text confirmation

Method	Endpoint	Authentication	Description	Request body	Response
DELETE	/users/delete/{user_id}	Administrator	Deletes a user by identifier.	None	Empty 204 response

Administration

Method	Endpoint	Authentication	Description	Request body	Response
GET	/admins	Administrator	Lists accounts with the administrator role.	None	UserOut []
POST	/admins/updateRole/{user_id}	Administrator	Changes a user's role.	{ "new_role": "USER" } or { "new_role": "ADMIN" }	Text confirmation
POST	/admins/update/is-active/{user_id}	Administrator	Activates or deactivates a user account.	{ "is_active": true }	Text confirmation
POST	/admins/wallet/balance/{user_id}	Administrator	Modifies a user's wallet balance.	{ "coins": 50 }	WalletResponse
GET	/admins/wallet/history?user_id={user_id}	Administrator	Retrieves a user's wallet history.	None	WalletResponse
GET	/admins/transaction/history?user_id={user_id}	Administrator	Retrieves a user's transaction history.	None	HistoryResponse
GET	/admins/reviews?user_id={user_id}	Administrator	Lists the reviews related to a user.	None	TransactionReviewOut []
GET	/admins/services?user_id={user_id}	Administrator	Lists the services published by a user.	None	ServiceOfferOut []

Method	Endpoint	Authentication	Description	Request body	Response
PATCH	/admins/services/{service_offer_id}/visibility	Administrator	Changes the visibility of a service offer.	{ "is_visible": true }	ServiceOfferOut
DELETE	/admins/services/{service_offer_id}	Administrator	Deletes a service offer from the administration area.	None	DeleteServiceOfferResponse
DELETE	/admins/reviews/{review_id}	Administrator	Deletes a review from the administration area.	None	Empty 204 response

Authentication

Method	Endpoint	Authentication	Description	Request body	Response
POST	/auth/token	No	Logs in and returns a JWT.	username , password in application/x- www-form- urlencoded format	{ "access_token": "...", "token_type": "bearer" }

Main Data Types

UserOut

Field	Type	Description
id	integer	User identifier.
email	string	User email address.
name	string or null	First name.
surname	string or null	Last name.
avatar_key	string or null	Selected avatar key.

Field	Type	Description
role	string	User role.
is_active	boolean	Account status.

PortalUserSummary

Field	Type	Description
id	integer	User identifier.
name	string	Name displayed in the sidebar.
role	string	USER or ADMIN .
email	string	Email address.
avatar_key	string or null	Selected avatar key.
pending_inbox_count	integer	Pending requests in the inbox.
pending_purchases_count	integer	Pending purchase updates.
pending_sales_count	integer	Pending sale updates.

ServiceOfferOut

Field	Type	Description
id	integer	Service identifier.
owner_id	integer or null	User who owns the service.
title	string	Service title.
description	string	Service description.
availability	string	Provider availability.
home_service	boolean	Indicates whether the service is performed at the requester's home.
address	string or null	Provider address when the service is not home based.
extra	string or null	Additional information.
price	integer	Price in coins.

Field	Type	Description
image_key	string	Image key used by the frontend.
owner_name	string	Provider display name.
is_visible	boolean	Indicates whether the service is visible in the catalogue.
overall_rating	float or null	Average service rating.

TransactionOut

Field	Type	Description
id	integer	Transaction identifier.
request_id	integer or null	Associated request.
chat_key	string or null	Associated chat thread key.
other_user_id	integer or null	Identifier of the other party.
unread_count	integer	Number of unread messages.
has_unseen_update	boolean	Indicates whether there is an unread update.
has_unseen_review	boolean	Indicates whether there is an unseen review.
type	string	Purchase or Sale .
service	string	Related service name.
other_user	string or null	Name of the other party.
date	string	Date in YYYY-MM-DD format.
address	string or null	Service address.
amount	integer	Coins transferred.
status	string	Transaction status.
clarification	string or null	Provider clarification when accepting a request.
reject_reason	string or null	Rejection reason.

ChatMessagesResponse

Field	Type	Description
messages	ChatMessageOut []	List of chat messages.

TransactionReviewOut

Field	Type	Description
id	integer	Review identifier.
transaction_id	integer	Reviewed transaction.
service	string or null	Related service.
reviewer_id	integer	User who writes the review.
reviewer_name	string or null	Name of the user who writes the review.
reviewed_user_id	integer	Reviewed user.
reviewed_user_name	string or null	Name of the reviewed user.
rating	integer	Rating between 1 and 5.
comment	string or null	Review comment.
created_at	string	Creation date.

User Guide

Installation and Local Execution

Frontend

Prerequisites

- Node.js 18 or higher.
- npm.

Steps

1. Open a terminal in the TimeBankFront/timebank-front/ folder.
2. Create a .env file with the backend URL:

```
VITE_API_URL=http://localhost:8000
```

3. Install the dependencies:

```
npm install
```

4. Run the development server:

```
npm run dev
```

The application usually starts at <http://localhost:5173>.

Backend

Prerequisites

- Python 3.10 or higher.
- MySQL server running locally or through Docker.

Steps

1. Open a terminal in the `TimeBankBack/` folder.
2. Create a virtual environment:

```
python -m venv .venv
```

3. Activate the virtual environment:

```
# Windows
.venv\Scripts\activate

# macOS/Linux
source .venv/bin/activate
```

4. Install the dependencies:

```
pip install -r requirements.txt
```

5. Configure the environment variables:

```
DB_HOST=localhost
DB_PORT=3306
DB_NAME=timebank
```



```
DB_USER=<database-user>
DB_PASSWORD=<database-password>
SECRET_KEY=<secret-key>
ALGORITHM=HS256
ACCESS_TOKEN_EXPIRE_MINUTES=60
STRIPE_SECRET_KEY=<stripe-secret-key>
STRIPE_WEBHOOK_SECRET=<stripe-webhook-signing-secret>
STRIPE_CURRENCY=eur
STRIPE_COIN_UNIT_AMOUNT_CENTS=100
FRONTEND_URL=http://localhost:5173
```

6. Run the API:

```
uvicorn app.main:app --reload
```

Regular User Guide

YouTube playlist with all user guide videos:

<https://www.youtube.com/playlist?list=PL5ISQmiP9ucbDI44ZIRndhnTX8tamhMLy>

Guide	Link
Create a user account	https://youtu.be/aqFwQxqLiRQ
Publish a service	https://youtu.be/6jkrCT_v5A0
Request a service	https://youtu.be/EcFJ-h6j50s
Delete a service	https://youtu.be/LmMkdHiBGOc
Accept a request in the inbox	https://youtu.be/8n2TKm9yS9s
Reject a request in the inbox	https://youtu.be/nKFCpO84tHo
Chat	https://youtu.be/6PqKpIIX1F0
Wallet	https://www.youtube.com/watch?v=0SwPBKf6YO4
Reviews	https://youtu.be/Q7kJC8p4IfI
Profile	https://youtu.be/CG-Kg5HE-jo

Administrator Guide

Administrator guide video:

<https://youtu.be/7uRWQPooUQk>